

ALI ASLANBAYLI

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Education

New York University

Masters of Science in Computer Engineering

GPA: 3.8 / 4.0

Sept. 2024 – May 2026

Brooklyn, NY

University of South Florida

Bachelors of Science in Computer Science

GPA: 3.7 / 4.0, Judy Genshaft Honors College

Aug. 2020 – May 2024

Tampa, FL

Technical Skills

Programming Languages: Python, C/C++, Go, Javascript, Typescript, PostgreSQL, Mojo

Libraries/Frameworks: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, FastAPI, React, Langchain, LLamaIndex

Developer Tools: Jira, Postman, Linux, Git, GCP, AWS, Docker, MLFlow, Ray, OpenShift

Experience

AI Engineer

July 2026 – Present

Providge Consulting

New York, NY

- Engineered multimodal document understanding systems capable of extracting and reasoning over complex PDFs containing tables, charts, forms, and scanned images using vision-language models and OCR pipelines.
- Developed autonomous AI agents that dynamically planned multi-step workflows, selected specialized tools, and synthesized information from heterogeneous enterprise data sources.
- Designed scalable document processing pipelines with layout analysis, metadata extraction, and asynchronous orchestration to support high-throughput enterprise AI applications.

GenAI Developer

Dec. 2024 – May 2026

NYU Research Technology Services

New York, NY

- Architected and deployed end-to-end Retrieval-Augmented Generation (RAG) pipelines using OpenWebUI and vector databases, directly improving model accuracy and reducing hallucinations for domain-specific queries.
- Engineered real-time data ingestion modules in Python that integrate Google Search APIs, increasing LLM context coverage by 40% and enabling the system to answer queries on live events.
- Built a user-facing agentic interface that automates complex data analysis, converting natural language directly into Pandas workflows to accelerate research operations.

Projects

Pelmen Coding Agent (<https://github.com/aslanbayli/pelmen>)

July 2026

- Built a TypeScript coding agent CLI from scratch using Bun, OpenRouter, Commander, and Ink/React, implementing a streaming agent loop with tool calling for file reads, surgical edits, file creation, and shell command execution.
- Architected a modular monorepo with AI, agent, and CLI packages, adding persistent session restore, model switching, context-window tracking with summarization, and a markdown-based skills system for reusable agent behaviors.

FireLens (<https://github.com/aslanbayli/firelens>)

July 2026

- Built a local-first hybrid code retrieval engine that indexes repositories into SQLite by extracting AST-based symbols, semantic code chunks, normalized embeddings, and file metadata with incremental reindexing, content hashing, and embedding reuse.
- Architected exact, fuzzy, and semantic search pipelines with query routing, MCP-compatible structured output, a Streamlit UI, and Mojo-accelerated kernels for fuzzy scoring, vector similarity, and top-k ranking on local codebases.

MeshNet (<https://github.com/aslanbayli/MeshNet>)

Dec. 2023

- Developed a Heterogeneous Graph Neural Network (GNN) using PyTorch Geometric to identify devices in mesh networks, designing a custom node schema (Devices, Events, Services) to model complex non-Euclidean streaming relationships.
- Optimized the training pipeline via hyperparameter sweeping and weighted loss functions to handle class imbalance, achieving 90% accuracy and surpassing Nielsen's benchmark for resolving erroneous MAC addresses.

Leadership

Society of Competitive Programmers - *Founder/President*

Aug. 2021 – Aug. 2023

- Hosted numerous competitive coding competitions, and mentored teams to attend the world-famous ICPC contests.

International Collegiate Programming Contest (ICPC) - *Silver Medalist*

March 2022

- Working in a team of two other students secured a silver medal at ICPC Southeast USA solving 5 out of the 7 problems.